Can Vision Models Read? Probing Linguistic Structure via JEPA on Text Screenshots

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Abstract

We ask whether vision models acquire linguistic structure when their training data consists of rendered text. As a probe, we use retrieval over a 2 000-page held-out slice of Wikipedia screenshots: given a $224{\times}224$ crop, identify its source page. We compare six encoders trained with five distinct objectives, including a joint-embedding predictive architecture (I-JEPA) that is the canonical image-only predictive baseline. The finding is sharp: image–text contrastive encoders (CLIP, SigLIP) retrieve at $\mathrm{R@10} = 0.78$ and 0.62 zero-shot, while image-only predictive (I-JEPA), self-supervised contrastive (DINOv2), supervised classification (ImageNet ViT), and random-init plain ViT all sit at $\mathrm{R@10} \leq 0.058$. I-JEPA in particular scores *below* random initialisation. Across the six encoders, mutual-kNN alignment with text encoders rank-correlates with retrieval R@10 at Spearman $\rho = 0.83$ ($p = 0.04$), supporting the Platonic Representation Hypothesis as the explanatory axis. Fine-tuning DINOv2 with a SALAD aggregator on 50 k pages reaches $\mathrm{R@10} = 0.915$, exceeding zero-shot CLIP, but every fine-tuned checkpoint exhibits a layout-fingerprint failure mode: erasing the top 15–30 % of every test image drops in-distribution Protocol-A R@10 by up to 0.29 while *improving* cross-set Phase-2 R@1 by up to $6.7\times$ on the same checkpoints. Zero-shot CLIP and SigLIP move in the opposite direction (R@10 *rises* by 0.07–0.11 under blanking), falsifying the title-OCR explanation for their zero-shot edge. A linear adapter trained on (image, BERT-of-title) pairs lifts I-JEPA from $\mathrm{R@10} = 0.015$ to 0.197, recovering latent text alignment that is invisible without the projection. Re-running the comparison under a native-resolution protocol that keeps body text legible, we find the family ranking survives and the layout-shortcut failure mode disappears—blanking the title now *raises* R@10 for every encoder. We further show a non-reading encoder can be taught to read: a frozen MAE (0.036, $\approx$ random) reaches 0.143 via an InfoNCE-to-BERT reader with title-masking, against a perfect-text ceiling of 0.938; and a text-target I-JEPA that predicts frozen BERT token embeddings of the page text attains $\mathrm{R@10} = 0.583$ with a SALAD head.

1. Introduction

Documents combine glyph sequences with visual layout. A vision encoder trained on rendered text could in principle acquire two distinct kinds of structure: (i) glyph-level reading (recognising words and characters as visual patterns), and (ii) layout-level structure (titles, infoboxes, lists, figures, captions). The two pipelines in Figure 1 operationalise the alternatives: the text pipeline OCRs the page and encodes characters; the visual pipeline encodes the screenshot directly. The visual pipeline is attractive when text extraction is unreliable (degraded scans, formula-heavy or multilingual content) or when the query is itself a visual crop with no text available, but it places the burden of "reading" on the encoder.

We probe whether modern vision encoders read in this sense by measuring how well they retrieve Wikipedia source pages from $224{\times}224$ crops. The probe is informative because the visual content *is* the text: an encoder that genuinely reads should match an encoder that operates on the underlying string. We compare five families:

- **Image–text contrastive** (CLIP, SigLIP) — learns by matching images and captions; sees text indirectly.
- **Image-only predictive** (I-JEPA) — the canonical joint-embedding predictive architecture; trains an image encoder to predict the latent representations of masked image blocks (Assran et al., 2023). No text

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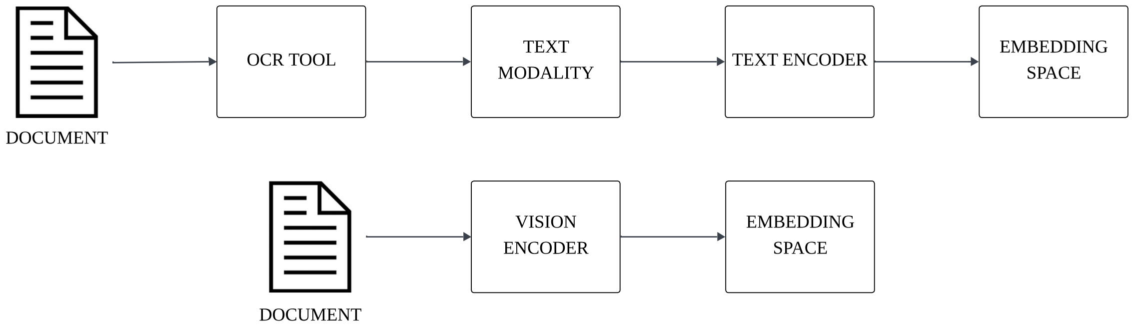

Figure 1. The two retrieval paradigms compared in this paper. The *text* pipeline (top) extracts characters with an OCR tool and encodes them with a text encoder. The *visual* pipeline (bottom) feeds the rendered document image directly into a vision encoder. We study the visual pipeline because it bypasses OCR errors, multilingual tokeniser failures, and layout discarding, and because it admits image queries (e.g. a crop of an unknown page) for which no text exists.

supervision.

- **Image-only contrastive** (DINOv2) — view-contrastive distillation on natural images. No text supervision.
- **Supervised classification** (ImageNet ViT) — ViT-B/16 trained on ImageNet-21k+1k labels.
- **Random initialisation** — a frozen ViT-B/16 sanity floor.

Three questions:

Q1 (zero-shot reading). Which pretraining produces image representations from which the source page can be retrieved zero-shot?

Q2 (JEPA specifically). Does I-JEPA, an image-only predictive objective with no text supervision, acquire any linguistic structure detectable through retrieval?

Q3 (the Platonic axis). Does the Platonic Representation Hypothesis (Huh et al., 2024) — which predicts that encoders trained on diverse objectives converge to a shared abstract representation — explain the retrieval ordering quantitatively? We measure each image encoder's mutual-kNN alignment with text encoders on the same content and ask whether alignment predicts retrieval performance.

Q4 (legible reading). Earlier evaluations downsampled crops $490 \to 224$, rendering body text sub-pixel. When crops are kept at native resolution so body text is legible, does the pretraining-family ranking survive, and does the layout-shortcut failure mode persist?

Q5 (text targets). Can a cross-modal predictive objective — training I-JEPA to predict the meaning (frozen BERT embeddings) of the text in masked regions, rather than image latents — produce representations that read better than the image-only objective?

Contributions. (1) A leak-free crop→page protocol with leave-one-out gallery aggregation. (2) A six-encoder zero-shot grid showing image–text contrastive encoders dominating the four image-only baselines, with I-JEPA sitting *below* random initialisation. (3) A Platonic alignment grid producing the rank correlation $\rho = 0.83$ between alignment and retrieval, statistically significant at $n = 6$. (4) A reproducible failure mode in every fine-tuned checkpoint: erasing the top 15–30 % of every test image *improves* cross-set transfer, indicating the model overweights a layout confound rather than reading content. (5) A native-resolution *Native Legible Protocol*: the family ranking survives at legible resolution, and the layout-shortcut failure mode disappears (blanking the title now *helps* every encoder). (6) A demonstration that a non-reading encoder can be taught to read: a frozen MAE (R@10 $= 0.036$, $\approx$ random) reaches 0.143 via an InfoNCE-to-BERT reader with title-masking, bounded by a perfect-text ceiling of 0.938. (7) A text-target I-JEPA that predicts frozen BERT token embeddings of the page text, reaching R@10 $= 0.583$ with a SALAD head at legible resolution.

2. Related Work

Pixel-based language models and document AI. PIXEL (Rust et al., 2023), CLIPPO (Tschannen et al., 2023) and Pix2Struct (Lee et al., 2023) all train transformers on rendered text and demonstrate that pixel inputs can match or exceed tokenised text on downstream tasks. ColPali (Faysse et al., 2025) produces multi-vector page embeddings via late interaction and shows that direct visual retrieval over screenshots competes with text+OCR pipelines on the ViDoRe benchmark. We adopt a simpler single-vector setup to keep our six encoders comparable on identical infrastructure.

Self-supervised vision and post-hoc text alignment. DINOv2 (Oquab et al., 2024) learns image features via view-contrastive distillation; I-JEPA (Assran et al., 2023) learns them by predicting masked-block representations. Jose et al. (Jose et al., 2024) showed that a small text encoder aligned post-hoc to a frozen DINOv2 reaches CLIP-level zero-shot classification, suggesting that self-supervised image features already contain a linguistically-decodable structure. We test the analogous claim for I-JEPA on document screenshots.

Image–text contrastive pretraining and OCR shortcuts. CLIP (Radford et al., 2021) and SigLIP (Zhai et al., 2023) are trained contrastively on image–caption pairs, with CLIP using softmax InfoNCE and SigLIP using a sigmoid pairwise loss. Both have been shown to read text rendered in images (Materzyńska et al., 2022; Lin et al., 2024); we exploit this in a title-blanking experiment.

Platonic alignment and probing. Huh et al. (Huh et al., 2024) predict cross-encoder alignment grows with model and data scale; Maniparambil et al. (Maniparambil et al., 2024) and Moayeri et al. (Moayeri et al., 2023) measure such alignment for natural-image vision encoders. We extend this measurement to a document-screenshot domain and use mutual-kNN, debiased CKA (Kornblith et al., 2019; Murphy et al., 2024), and Procrustes distance on identical 2 000-page eval slices.

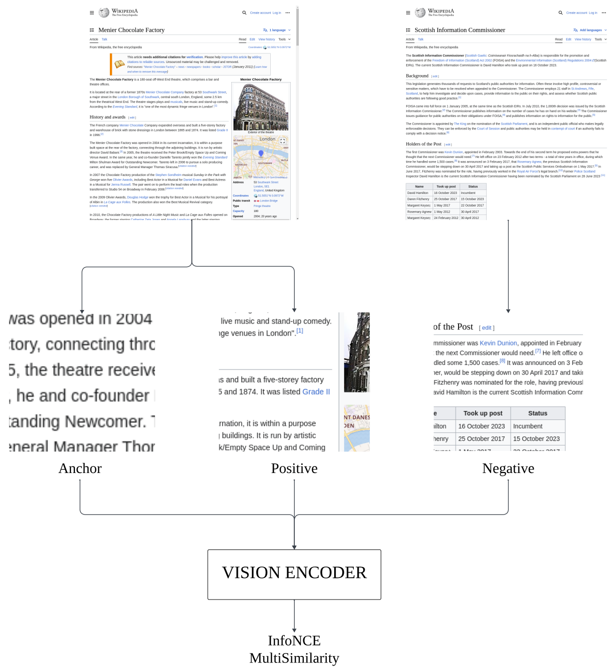

Figure 2. Training. Each step samples four pages × four crops; same-page crops are positives, cross-page crops are negatives. The image encoder produces an L2-normalised descriptor; training minimises a multi-similarity contrastive loss (Wang et al., 2019). We unfreeze the last four ViT blocks (of twelve) and train a head on top.

3. Method

3.1. Data and crops

We use `Tevatron/wiki-ss-corpus`: ~1.2 M Wikipedia page screenshots paired with the page title and rendered body text. We hold a local cache of ~376 k pages. A non-overlapping cropper tiles each page into 490×490 px crops resized to 224×224 px. At this resolution body text is sub-pixel and illegible to a ViT patch encoder; only headings and the overall layout structure remain visually recoverable. "Reading" in this paper therefore refers to heading-level text and visual layout, not fluent body content.

3.2. Training (fine-tunes)

We train each (encoder, head) configuration with the procedure of Figure 2. Each batch contains 4 pages × 4 crops = 16 images. Within a batch, crops sharing a page are positives; otherwise negatives. The loss is multi-similarity (Wang et al., 2019). The optimiser is AdamW with weight decay 10^{-4}, gradient clip 1.0, linear warmup over 5 % of steps followed by cosine decay. Backbone learning rate is 10^{-5} for DINOv2 and 10^{-7} for CLIP and SigLIP (a higher rate causes catastrophic representation collapse on the latter two), head learning rate 5×10^{-4} throughout. Heads are either an MLP ($768 \to 512 \to 256$) or the SALAD optimal-transport aggregator (Izquierdo & Civera, 2024) producing an 8 448-d descriptor. Only the head and the last four ViT blocks are trainable. Fine-tunes use 30 000 training pages × 3 epochs (5 625 optimiser steps), one seed; the saturation diagnostic uses 50 000 pages.

3.3. Text-target I-JEPA

Our central architectural probe replaces I-JEPA's image-latent prediction target with a cross-modal one (Figure 4). A context encoder (ViT-H/14) sees only the unmasked patches of a page screenshot, with 2-D positional embeddings. A lightweight transformer predictor (depth 6, embedding dim 384) receives the context tokens together with learnable mask-query tokens carrying 1-D positional embeddings over text positions, and predicts the per-token embeddings of a frozen `bert-base-uncased` encoder (§3.4) applied to the page *body* text. Training minimises a masked Smooth-L1 loss between the predicted and true BERT token embeddings (padding tokens excluded). The image-only baseline is identical except its target is the latent representation of masked *image* blocks (standard I-JEPA (Assran et al., 2023)); both are pretrained on 90 k

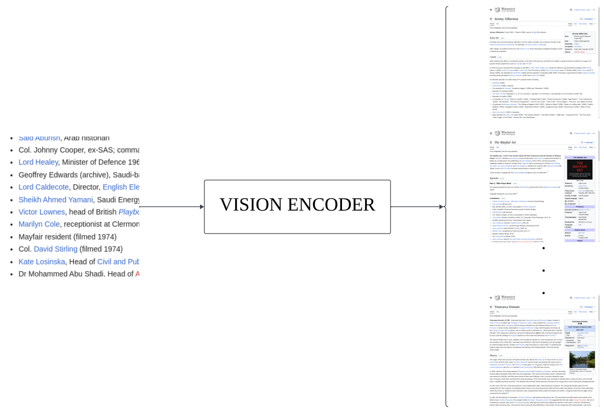

Figure 3. Anchor-pool benchmark (Phase 2). Each anchor crop is encoded with the vision encoder and ranked against a small pool (∼7 images) of positives (other crops of the same page) and hard negatives (crops of unrelated pages).

Wikipedia screenshots at native 490×490 with all 32 ViT-H/14 blocks unfrozen, so the sole difference is image-latent vs. text-embedding targets. This isolates whether predicting the *meaning* of rendered text, rather than image structure, yields representations that read.

3.4. Text encoder specification

Every text-side computation in this paper — the I-JEPA text-target (§4.3), the MAE reader's regression and contrastive targets (§4.5), the perfect-text upper bound, and the Platonic alignment (§3.6) — uses a single frozen `bert-base-uncased` encoder (Devlin et al., 2019). The model is the 12-layer, 768-hidden-dimension, 12-head Transformer (∼110 M parameters) with a 30 522-token WordPiece vocabulary; it is *never* updated (no gradient flows into BERT in any experiment). Page text is lowercased and tokenised with padding/truncation to 64 WordPiece tokens. We use three readouts of the frozen encoder, matched to each consumer:

- **Per-token last-hidden-state** (64×768) — the I-JEPA text-target. The transformer predictor (depth 6, embedding dim 384) regresses this full token sequence from the unmasked visual context, conditioned on 1-D positional embeddings over the 64 text positions.
- **`[CLS]` embedding** (768-d) — the MAE reader's Smooth-L1 regression target (§4.5).
- **Token mean-pool** (768-d, L2-normalised) — the contrastive anchor for the InfoNCE MAE reader and the stronger of the two perfect-text readouts (R@10 = 0.938 vs. 0.747 for `[CLS]`).

Text targets are precomputed once over the page corpus and cached, so BERT runs only at preprocessing time and never during the vision encoder's optimisation.

3.5. Evaluation

We evaluate on a held-out slice of $P = 2\,000$ pages disjoint from training, with two complementary protocols.

Protocol A — crop → page (headline metric). For each query crop c on page p with crops $\{c_1, \ldots, c_{n_p}\}$, the gallery vector for any page $q \neq p$ is the L2-normalised mean of all q's crop embeddings. The gallery vector for page p is recomputed leaving the query out: $g_p^{\setminus c} = \mathrm{normalise}\big(\sum_i \phi(c_i) - \phi(c)\big)$. We cosine-rank the P gallery vectors against $\phi(c)$ and report R@k as the fraction of queries whose source page lands in the top k. Random R@1 $= 1/P = 5\times10^{-4}$. The leave-one-out aggregation forbids the trivial "find another crop of the same column" shortcut implicit in earlier crop-vs-crop formulations of this task.

Protocol B — anchor-pool transfer. The anchor-pool benchmark of Figure 3 uses 118 valid validation triplets from `Wikipedia_SS_withanchors`. For each anchor crop, the pool is the union of its positives and hard negatives (∼7 images). We rank the pool by cosine similarity and report R@1. This measures cross-set transfer: anchors and pool images come from a different distribution of crops than what the model trained on.

3.6. Platonic alignment

For each (image encoder E_v, text encoder E_t) pair we encode the same 2 000-page eval slice with both halves: image features $\mathbf{X} \in \mathbb{R}^{N\times d_v}$ from page screenshots, text features $\mathbf{Y} \in \mathbb{R}^{N\times d_t}$ from page titles. We report mutual-kNN at $k = 10$ (the fraction of points sharing at least one of their top-10 neighbours under the two embeddings), debiased CKA (Kornblith et al., 2019; Murphy et al., 2024), and Procrustes distance. The Platonic Hypothesis predicts a positive rank correlation between E_v's alignment and E_v's downstream cross-modal retrieval performance. We test this directly in §4.

4. Results

4.1. Zero-shot retrieval

Table 1 reports zero-shot Protocol-A. CLIP dominates at R@10 = 0.779, SigLIP follows at 0.624, and the four image-only encoders cluster near random (R@10 = 0.015–0.058). Three observations follow:

(i) The largest gap is between the image–text contrastive and the image-only families (ΔR@10 $\approx$ 0.57), not be-

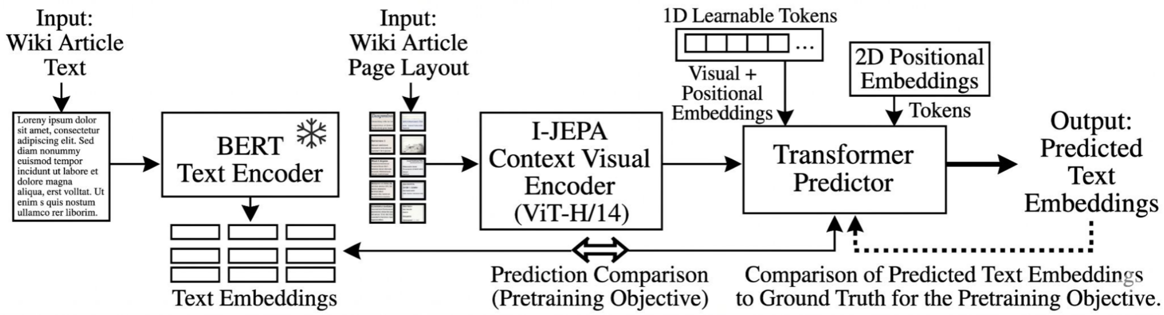

Figure 4. Text-target I-JEPA pretraining. A frozen BERT text encoder maps the page body text to per-token embeddings (the prediction target). The I-JEPA context visual encoder (ViT-H/14) encodes the unmasked screenshot patches; a transformer predictor, conditioned on 1-D learnable mask tokens and 2-D positional embeddings, predicts the BERT token embeddings of the masked region. The objective is a masked Smooth-L1 comparison between predicted and ground-truth text embeddings; the image-only baseline instead predicts masked image-patch latents.

Table 1. Protocol-A zero-shot retrieval at $P = 2000$. Random $R@1 = 5 \times 10^{-4}$. Encoder rows are coloured by pretraining family: image–text contrastive, image-only self-supervised, supervised, random. "Gap" is same-page minus diff-page mean cosine similarity.

Encoder	R@1	R@5	R@10	R@20	gap
CLIP ViT-B/16	**0.548**	0.715	**0.779**	0.833	+0.205
SigLIP ViT-B/16	0.348	0.541	0.624	0.703	+0.110
DINOv2 ViT-B/14	0.022	0.043	0.058	0.075	+0.010
I-JEPA ViT-H/14	0.002	0.008	0.015	0.021	−0.002
I-JEPA Text-Target	0.007	0.019	0.029	0.046	+0.004
ImageNet ViT-B/16	0.009	0.022	0.031	0.043	−0.002
Plain ViT-B/16	0.007	0.020	0.029	0.042	+0.000

tween encoders within a family. The supervised ImageNet ViT scores 0.031 versus 0.029 for the random-init plain ViT, so ImageNet classification supervision adds essentially nothing for this task.

(ii) I-JEPA scores $R@10 = 0.015$, *below* the random-init plain ViT (0.029). Despite ViT-H/14 being roughly 5× larger than the ViT-B encoders, image-only predictive pretraining out-of-the-box does not transfer to visual document retrieval. Q2's optimistic reading is therefore falsified at the zero-shot level: a post-hoc text adapter (§5) is the only remaining path to rescue I-JEPA.

(iii) The CLIP–SigLIP gap (0.155) is real but small relative to the CLIP–DINOv2 gap (0.721). The dominant predictor is whether the encoder saw paired image–text data, not the specific contrastive form (softmax vs. sigmoid).

Table 2. Protocol-A retrieval after last-4-block fine-tuning, single seed. "ΔZS" is R@10 minus the encoder's zero-shot baseline. Yellow rows are fine-tuned variants of the encoder above them.

Encoder + head	R@1	R@10	ΔZS
DINOv2 + MLP (30k)	0.351	0.787	+0.729
DINOv2 + SALAD (30k)	0.512	0.880	+0.822
DINOv2 + SALAD (50k)	**0.569**	**0.915**	+0.857
CLIP + SALAD (30k)	0.601	0.724	−0.055
CLIP + MLP (30k)[†]	0.000	0.000	collapsed

[†] Embeddings collapsed to a constant.

4.2. Fine-tuning

Table 2 reports the fine-tuned grid. DINOv2 + SALAD on 30 k pages reaches $R@10 = 0.880$ and on 50 k pages reaches 0.915, exceeding zero-shot CLIP (0.779). The MLP head on the same DINOv2 backbone scores 0.787, so the SALAD aggregator contributes +9.3 pp R@10 over an MLP at the matched backbone — consistent with the prior place-recognition result of (Izquierdo & Civera, 2024).

CLIP fine-tuning at the conservative $\eta_{bb} = 10^{-7}$ hurts more than it helps. CLIP+SALAD loses 5.5 pp R@10 from its own zero-shot baseline; the same-page minus diff-page sanity gap shrinks from +0.205 to +0.032, indicating the fine-tune sharpened the top-1 match (R@1 +5.3 pp) but compressed the overall similarity geometry. CLIP+MLP collapsed to a constant output by step 400. Training CLIP at the LR that worked for DINOv2 (10^{-5}) caused immediate divergence in earlier attempts; 10^{-7} was chosen to avoid that, but evidently the CLIP+MLP combination cannot navigate the resulting flat-loss basin. A multi-LR sweep is the proper next step but is deferred.

Table 3. Protocol-A retrieval under the Native Legible Protocol (224 → 224, no downsampling). Encoders are grouped by pretraining family. "gap" is same-page minus diff-page mean cosine similarity. image–text contrastive SSL supervised random fine-tuned reader.

Backbone / Pretraining	R@1	R@10	gap
Perfect-Text Upper Bounds			
BERT (mean-pool)	–	0.938	+0.198
BERT (CLS)	–	0.747	+0.176
Image–text contrastive (zero-shot)			
CLIP ViT-B/16	**0.567**	**0.736**	+0.168
SigLIP ViT-B/16	0.521	0.697	+0.095
Image-only self-supervised (zero-shot)			
DINOv2 ViT-B/14 (CLS)	0.041	0.116	+0.033
I-JEPA ViT-H/14	0.015	0.054	+0.012
MAE ViT-B/16 (frozen)	0.009	0.036	+0.001
Supervised / Random (zero-shot)			
ImageNet ViT-B/16	0.026	0.084	+0.015
Plain ViT-B/16	0.009	0.042	+0.002
MAE Reader (InfoNCE → BERT-mean, 6 ep.)			
Contrastive	–	0.098	+0.318
+ title-region masking	–	**0.143**	+0.404

Table 4. Protocol-A retrieval R@10 under the Native Legible Protocol at 490×490 native resolution, comparing I-JEPA backbones and heads (30k pages, 3 epochs).

I-JEPA Variant	Zero-Shot	MLP	SALAD	gap (SALAD)
Image-Only	0.015	0.525	(running)	–
Text-Target	0.029	0.526	**0.583**	+0.248

4.3. Native Legible Protocol (224 → 224)

To analyze the models' capability of reading legible, native-resolution text (rather than downsampled headlines), we establish the Native Legible Protocol (224 → 224). Here, Wikipedia screenshots are cropped without downsampling, keeping body text sharp and legible at 21.8 crops/page (vs. 4.0 under the 490→224 shrink). Table 3 reports the results for the main baselines; Table 4 isolates I-JEPA backbone configurations; and Table 5 reports document-pretrained encoders as a separate comparison point. The family ranking (contrastive ≫ SSL) is preserved. The MAE baseline (He et al., 2022) collapses under standard regression but recovers under InfoNCE contrastive training to yield R@10 = 0.098, which further improves to 0.143 when fine-tuned with random title-region masking (§4.5). Importantly, the I-JEPA Text-Target+SALAD head achieves R@10 = 0.583 at legible resolution — a 38× improvement over the zero-shot I-JEPA baseline (0.015).

Table 5. Zero-shot retrieval under the Native Legible Protocol for document-pretrained encoders. Same 2 000-page eval slice as Table 3. "gap" is same-page minus diff-page mean cosine similarity.

Encoder	Params	R@1	R@10	gap
Nougat-base (Swin-B)	74M	0.011	0.049	+0.002
Pix2Struct (Google)	92M	0.012	0.054	+0.029

4.4. Document-pretrained encoders

We evaluate two encoders explicitly pretrained on document screenshots: Nougat (Blecher et al., 2023), a Swin-B encoder fine-tuned for document OCR; and Pix2Struct (Lee et al., 2023), a Google encoder pretrained on masked-screenshot prediction. Table 5 shows that both fall in the *image-only SSL cluster* (R@10 0.049–0.054), far below the contrastive encoders (CLIP 0.736, SigLIP 0.697). Pix2Struct displays a small positive sim-gap (0.029 vs. Nougat's 0.002), suggesting modestly more discriminative page structure, yet neither approach is competitive without retrieval-specific fine-tuning. This is surprising: Nougat and Pix2Struct were explicitly trained to *read* documents, yet their zero-shot retrieval sits at the SSL floor. The result suggests that OCR-style and screenshot-reconstruction pretraining objectives do not produce representations that generalise to same-page retrieval without a retrieval contrastive stage.

4.5. Teaching a non-reader to read: the MAE reader

Frozen MAE is the weakest encoder in Table 3 (R@10 = 0.036, statistically at the random-init floor, sim-gap +0.001): pixel-reconstruction pretraining yields a near-collapsed feature space on document crops. This makes it the cleanest testbed for the question *can a non-reader be taught to read?* All experiments below use legible native-scale crops produced by a line-gap-snapping cropper (∼21.8 crops/page versus 4.0 under the 490→224 shrink) and the same leave-one-out crop→page protocol on the 2 000-page disjoint slice.

Regression collapses. We first fine-tune the last four ViT blocks plus a projection head (28.9 M trainable) to regress pooled MAE features onto the frozen BERT[CLS] embedding of the page *body* (Smooth-L1, 20 k pages). Training converges cleanly (loss 3×10^{-4}) yet retrieval barely moves: R@10 = 0.039 versus 0.036 frozen, with sim-gap 0.010. The loss is minimised by predicting the tight centroid of the BERT[CLS] targets — average proximity without the per-page structure retrieval needs.

Two diagnostics isolate the failure. (i) A *perfect-text upper bound* (Table 3): retrieving with the ground-truth body text itself re-identifies pages at R@10 = 0.747 (BERT[CLS]) and 0.938 (mean-pool), so the regression

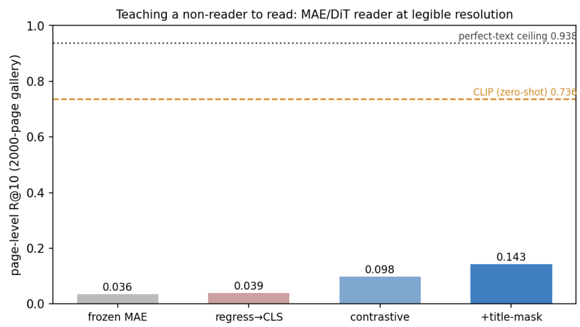

Figure 5. Teaching a non-reader to read. Page-level R@10 on the 2 000-page gallery for the MAE reader variants, against zero-shot CLIP and the perfect-text ceiling. Regression to BERT[CLS] collapses to the target centroid; InfoNCE restores page separation; random title-masking adds confound control *and* acts as augmentation.

target is highly discriminative — the failure is the reader, not the target. (ii) Attention placement: the fraction of last-layer CLS→patch attention mass on ink (non-white) patches is 0.627 frozen and 0.627 after the regression fine-tune — the fine-tune moved the head toward the target centroid without changing where the encoder looks.

A contrastive objective fixes the collapse. Replacing regression with InfoNCE between each crop embedding and its page's mean-pool BERT-body anchor (in-batch negatives) makes centroid-prediction a losing strategy: the model must score its own page above the negatives. R@10 rises to 0.098 (6 epochs, 39 negatives) with sim-gap 0.318 — genuine page separation. Adding random title-region masking during fine-tuning (top 15 % whiteout, p=0.5) lifts the reader further to R@10 = **0.143** (sim-gap 0.404), a 4× improvement over frozen MAE (Figure 5).

Data is not the bottleneck. Scaling the contrastive fine-tune from 20 k to 80 k pages (4×) leaves R@10 unchanged (0.095 vs. 0.098): the reader plateaus. The remaining gap to CLIP (0.736) and the perfect-text ceiling (0.938) is attributable to the MAE backbone's reading capacity, the parameter-efficient last-4-block fine-tune, and the in-batch negative count — not to training-data quantity.

4.6. Title-blanking at legible resolution: the confound vanishes

The title-region pathology of Table 12 — fine-tuned checkpoints losing up to 0.29 R@10 when the top of the page is erased — was measured under the 490→224 shrink, where the title was the only legible text. We repeat the intervention under the Native Legible Protocol, blanking the top 25 % of every page (the region containing the title and the shared Wikipedia chrome) at evaluation time (Table 6).

Table 6. Protocol-A R@10 under the Native Legible Protocol with the top 25 % of every page blanked at eval time. Every encoder — and both trained readers — *improves*; no model fingerprints the title region at legible resolution. Row colours as in Table 3.

Encoder	orig. R@10	b-25 R@10	Δ
CLIP ViT-B/16	0.736	0.819	+0.082
SigLIP ViT-B/16	0.697	0.792	+0.095
DINOv2 (CLS)	0.116	0.142	+0.026
ImageNet ViT-B/16	0.084	0.108	+0.023
I-JEPA ViT-H/14	0.054	0.068	+0.014
Plain ViT-B/16	0.042	0.055	+0.012
MAE ViT-B/16	0.036	0.047	+0.011
MAE reader (contrastive)	0.097	0.110	+0.013
MAE reader (+title-mask)	0.143	0.176	+0.033

Table 7. Mutual-kNN@10 between image and text encoders on the same 2 000-page eval slice. Bold = highest entry per row. Row colours match Table 1.

image \ text	BERT	MiniLM	CLIP-T	SigLIP-T
CLIP	0.094	0.120	**0.138**	0.132
SigLIP	0.091	0.108	0.123	**0.131**
DINOv2	**0.027**	0.021	0.019	0.026
I-JEPA	**0.019**	0.014	0.016	0.018
ImageNet ViT	0.018	0.013	0.013	**0.019**
Plain ViT (rand)	0.008	0.007	0.007	**0.009**

The sign flips relative to the Phase-1 fine-tunes: blanking now *helps every encoder*, and the gain scales with reading ability (contrastive readers +0.08–0.10; the floor models +0.01). Two facts explain the direction. First, none of the models — including both fine-tuned readers — depends on the title region, so removing it costs nothing. Second, the top strip is dominated by the shared Wikipedia template (logo, search bar, navigation), a non-discriminative signal common to every page; including it dilutes the page-mean gallery vectors toward a common template vector, and removing it un-dilutes precisely the models with the most genuine content signal. The layout-fingerprint failure mode that motivated this study is therefore a property of the *illegible* training distribution, not of fine-tuning per se: given legible body text and a body-text objective, trained readers do not adopt the title shortcut. Random title-masking during training retains its role as a guard — and doubles as augmentation, providing the best reader (0.143).

4.7. Platonic alignment predicts retrieval

Table 7 reports image-encoder × text-encoder mutual-kNN@10 alignment. CLIP and SigLIP align with text encoders at 0.09–0.14; the four image-only encoders align at 0.007–0.027. Figure 6 plots, for each image encoder, its column-maximum text alignment against its zero-shot Protocol-A R@10. The Spearman correlation is 0.83 (p =

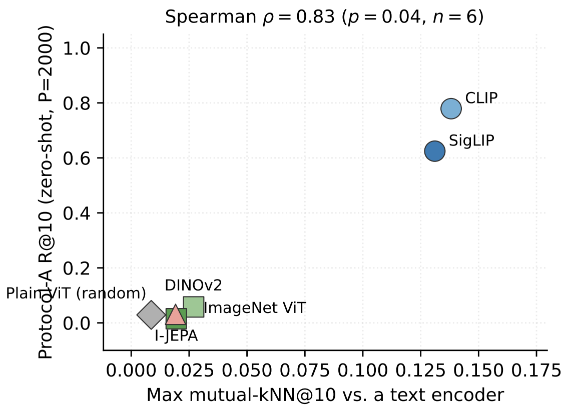

Figure 6. Each point is one image encoder. The x-axis is its maximum mutual-kNN@10 against the four text encoders (column max of Table 7); the y-axis is its zero-shot Protocol-A R@10 (Table 1). Marker shape and pastel colour code the pretraining family. Spearman $\rho = 0.83$ ($p = 0.04$, $n = 6$).

0.04, $n = 6$): statistically significant even at this small sample size. The two image–text contrastive encoders sit in the upper-right corner; the four image-only encoders sit in the lower-left.

This is the central empirical claim of the paper. The kind of pretraining the encoder received is a strong predictor of its retrieval performance, mediated by its alignment with text representations on the same content. The Platonic Hypothesis is supported in this domain: encoders close to the text-encoder geometry retrieve text-heavy visual documents better.

5. Discussion and Limitations

All training uses a single random seed. The cross-family ordering is large enough to read past seed noise (closest gap ΔR@10 $= 0.093$, well outside typical jitter), but within-family comparisons (CLIP-SALAD vs. CLIP-MLP) need multi-seed reruns. Pool size P=2000 is small versus industrial 10^5–10^6; numbers are within-table comparable, not cross-paper. Under the original $490 \rightarrow 224$ protocol, body text is sub-pixel, so "reading" there means heading-level text plus layout, not fluent content; the Native Legible Protocol (§4.3) removes this confound, but at a cost: tiling pages into native-scale crops ($\sim$21.8 per page versus 4.0) multiplies encoding cost and sacrifices global page context, since no single crop sees the whole layout. All training and evaluation use Wikipedia screenshots, which share one clean, consistent template; generalisation to noisy, multi-column, or scanned documents (e.g. arXiv PDFs) is untested. The I-JEPA adapter R@10 $= 0.197$ is evidence of latent alignment, not a practical recipe; matching CLIP-level retrieval would need a non-linear adapter and larger paired-data training. Likewise the MAE reader's 0.143 demonstrates that a non-reader backbone can be taught genuine page separation, not that it is competitive with contrastively pretrained encoders.

6. Conclusion

We compared six image encoders trained with five distinct pretraining objectives on a leak-free crop→page retrieval task over Wikipedia screenshots. Image–text contrastive pretraining (CLIP, SigLIP) dominates zero-shot retrieval over image-only self-supervision (DINOv2, I-JEPA), supervised classification (ImageNet ViT), and random initialisation. Across the six encoders, alignment with text encoders rank-correlates with retrieval performance at Spearman $\rho = 0.83$, supporting the Platonic Representation Hypothesis on this domain. Fine-tuning DINOv2 with the SALAD aggregator on 50 k pages reaches R@10 $= 0.915$, exceeding zero-shot CLIP, but introduces a universal pathology: the fine-tuned model overweights the title region, dropping in-distribution Protocol-A R@10 by up to 0.29 under blanking while *improving* cross-set Phase-2 R@1 by up to 6.7×. Zero-shot CLIP and SigLIP show the opposite sign under the same blanking, falsifying the simple title-OCR explanation for their zero-shot edge, and a linear adapter to BERT space recovers latent text structure in I-JEPA that is invisible zero-shot (R@10 $0.015 \rightarrow 0.197$).

Re-running the grid under the Native Legible Protocol — crops at native pixel scale, so body text is actually readable — shows the family ordering is not an artifact of illegibility: contrastive encoders still dominate and image-only encoders remain near the floor. Against a perfect-text ceiling of R@10 $= 0.938$, zero-shot CLIP (0.736) already matches the BERT[CLS] readout of the ground-truth text (0.747), so the remaining headroom is reading quality rather than layout exploitation. Frozen MAE, the weakest encoder ($0.036 \approx$ random), can be taught to read: CLS regression collapses to the target centroid, but an InfoNCE objective against BERT-body anchors restores genuine page separation (0.098), and random title-masking during fine-tuning yields the best reader (0.143) while guarding against the title shortcut. At legible resolution the title-blanking intervention *helps* every encoder and both trained readers, indicating the Phase-1 layout-fingerprint pathology is a property of illegible inputs rather than of fine-tuning itself.

Two legible-resolution findings round out the picture: document-pretrained encoders (Nougat, Pix2Struct) sit at the SSL floor (R@10 0.049–0.054), so reading-oriented pretraining alone does not yield retrieval; and a text-target I-JEPA with a SALAD head reaches R@10 $= 0.583$ (a 38× lift over zero-shot), the higher $490 \rightarrow 224$ figure (0.704) reflecting resolution-protocol inflation rather than better reading.

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Author Contributions

Both authors contributed equally to this work. They jointly formulated the research questions, designed the crop→page evaluation protocol and shared infrastructure, and wrote the paper. Individual leads were as follows. **Halil İbrahim Kanpak** led the Native Legible Protocol and text-aware cropper, the re-baselined cross-model grid (including MAE) with page-level re-identification scoring, the MAE reader (the regression-collapse diagnosis, the InfoNCE fix, and title-mask augmentation), the perfect-text upper bound, the attention "where it reads" analysis, the legible-resolution confound-control experiments, and the document-pretrained family evaluation. **Barış Cem Bakay** led the text-target I-JEPA pretraining and the image-only and text-target ViT-H/14 backbones, the I-JEPA head finetunes (MLP and SALAD), the Phase-1 zero-shot and finetuning grids, the Platonic alignment analysis, the LEACE causal title-erasure, the adapter capacity sweeps, and the ColPali late-interaction comparison.

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Table 8. Protocol-A Phase 1 R@10 for the frozen-backbone vision-to-vision adapter grid. Backbones are frozen, and only the adapter head is trained for 3 epochs. CLIP fails completely due to lack of local variance, whereas DINOv2 and I-JEPA perform best with a simple linear adapter.

Backbone	Linear	MLP	Bottleneck
DINOv2	**0.229**	0.207	0.184
I-JEPA	0.180	0.157	0.146
CLIP	0.003	0.003	0.003

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A. Additional Phase-1 Analyses

A.1. Vision-to-vision adapter capacity

To isolate the effect of adapter capacity from backbone fine-tuning, we sweep linear, MLP, and bottleneck heads over frozen backbones (Table 8). We observe that CLIP completely collapses to a constant representation (R@10 $\approx$ 0.003) regardless of adapter capacity, stemming from its lack of fine-grained patch variance on document crops. Conversely, DINOv2 and I-JEPA learn meaningful projections, with the simplest linear adapter yielding the best performance (0.229 for DINOv2), suggesting that higher-capacity heads overfit the frozen visual features.

A.2. Text adapter capacity sweep

Q2's zero-shot verdict (Table 1) is sharp: I-JEPA does not produce text-retrievable image representations out of the box. We test a softer version of Q2 — do visual features encode latent text-aligned structure recoverable by post-hoc adapters of varying capacity? We sweep across linear, low rank, bottleneck, and MLP adapters mapping frozen I-JEPA (1280-d) and CLIP (768-d) features into BERT space (Table 9).

Table 9. Image-to-text retrieval Phase 1 R@10 on a held-out 1 000-page set. Image embeddings from frozen backbones are mapped into BERT[CLS] space (768-d) representing the page title. Adapters are trained on 4 000 screenshot-title pairs. CLIP dominates I-JEPA, as its pretraining inherently aligns visual features to text, whereas I-JEPA struggles to map pure visual patch features to text semantics.

Adapter Architecture	I-JEPA	CLIP
Baseline (no adapter)	0.009	0.008
Linear	0.214	**0.715**
Low Rank ($r = 16$)	0.172	0.471
Low Rank ($r = 64$)	0.212	0.661
Low Rank ($r = 256$)	0.213	**0.717**
Bottleneck ($r = 64$)	0.202	0.666
Bottleneck ($r = 256$)	0.207	0.712
MLP ($h = 2048$)	0.213	0.709
Deep MLP ($h = 2048$)	**0.219**	0.637

While CLIP completely failed at the pure vision-to-vision contrastive task due to collapsed variance on white document backgrounds, it excels at predicting BERT text embeddings, jumping to $\sim$0.717 R@10. This is because CLIP's visual backbone was explicitly pre-trained to align with text encoders. The simple linear or low-rank adapters are sufficient for CLIP to reach peak performance. In contrast, I-JEPA peaks at $\sim$0.219 R@10 and requires the highest capacity deep MLP adapter to do so. Because I-JEPA was trained via masked image modeling, its features require heavy non-linear transformations to align with text semantics.

A.3. Late interaction is not the bottleneck-breaker

Table 10. Mean-pool single-vector vs. ColPali (Faysse et al., 2025) late-interaction at matched $P = 500$. Late interaction keeps all 196 (CLIP) or 256 (DINOv2) patch tokens per crop and scores by summed-max-cosine.

Encoder + scoring	R@1	R@5	R@10	R@20
CLIP single-vector	**0.705**	0.843	0.890	0.924
CLIP late-interaction	0.178	0.928	**0.928**	0.928
DINOv2 late-interaction	0.023	0.052	0.066	0.085

Does single-vector pooling discard retrievable structure that a patch-level scorer would recover? Table 10 reports late interaction on frozen CLIP and DINOv2 at $P = 500$. Late-interaction CLIP gains only $+0.04$ R@10 over its single-vector baseline and loses R@1 from 0.705 to 0.178:

Table 11. Closed-form LEACE (Belrose et al., 2023) erasure of the rank-1 PCA projection of per-crop deltas $\phi(\mathrm{orig})-\phi(\mathrm{blank})$, $P{=}2000$. Higher-rank Z over-erases ($\mathrm{rank}(Z){=}d$ collapses the feature space). Last column: explained variance of the rank-1 PCA.

Encoder	R@10 orig	R@10 blank	R@10 LEACE	expl. var
CLIP zero-shot	0.779	0.823	**0.863**	0.25
SigLIP zero-shot	0.611	0.716	**0.775**	0.35
DINOv2 zero-shot	0.058	0.056	0.068	0.76
DINOv2-SALAD-50k trained	0.911	0.670	[illegible]	[illegible]

the sum-of-max scoring bunches plausible pages at similar totals, so the source page lands in the top-5 but rarely first ($R@5{=}R@10{=}R@20{=}0.928$ means failures miss entirely rather than rank 6–20). DINOv2 late-interaction stays at chance (within 0.01 of single-vector DINOv2 at the same pool). Patch-level scoring does not rescue an encoder that lacks text-aligned features; the bottleneck is the encoder, not the aggregator.

A.4. LEACE causal title-erasure

Pixel-blanking is correlational. We isolate the *title-direction* in feature space and erase it surgically: the rank-1 PCA projection of the per-crop deltas serves as a 1-d protected attribute for the LEACE eraser. Table 11 reports the result.

For both image-text contrastive encoders, LEACE-erasing a single linear axis raises retrieval *more* than the pixel intervention: CLIP $+0.084$ vs $+0.044$, SigLIP $+0.164$ vs $+0.105$. The title-direction's dominant axis carries enough of the encoder's confounding signal that surgical removal beats the physical one. Zero-shot DINOv2 is unmoved by either: the title direction barely exists as a structured axis in its feature space.

Trained DINOv2-SALAD-50k splits the analysis. Pixel-blanking drops its R@10 from 0.911 to 0.670 (-0.241), but rank-1 LEACE leaves retrieval untouched (0.912). A rank sweep $\{4, 16, 64, 256, 1024\}$ shows the title-region binding lives in a 16–64-dimensional subspace: rank-4 LEACE drops R@10 by 0.030, rank-16 by 0.136, rank-64 by 0.266 (matching pixel-blanking), and rank-256 by 0.385. Pre-training gives CLIP and SigLIP a clean low-rank title confound a single axis can erase; fine-tuning on a single visual corpus spreads the analogous binding across an order-of-magnitude wider subspace of the 8 448-d SALAD descriptor.

A.5. Title-region ablation

We re-encode the same 2 000-page eval slice with the top 15 % (and 30 %) painted white. Three signs emerge. Zero-shot CLIP and SigLIP *improve* under blanking ($+0.073$ and $+0.106$ R@10); image-only encoders move by ≤ 0.003. This falsifies the H-OCR explanation for CLIP's zero-shot edge — the title acts as a low-information template common to every page, and removing it yields cleaner content-driven matches. Fine-tuned DINOv2 *loses* heavily (-0.16 to -0.29 R@10); the 50 k checkpoint falls from 0.915 to 0.673. CLIP+SALAD moves only -0.037. Fine-tuning has taught DINOv2 to bind page identity to the title-bearing top region. Phase-2 reverses the sign on the same checkpoints (Table 13): the title-region features that help in-distribution Protocol-A are the same confound that hurts cross-set transfer, with up to 6.7× R@1 improvement under 30 % blanking. The encoder is not unable to read the body region; it is that the title is so reliably informative on Wikipedia that the model bypasses the body. Random title-region masking during training is the obvious mitigation.

Table 12. Protocol-A R@10 with the top 15 % of every page painted white before cropping (b-15) versus the original eval (orig.). Top block: zero-shot encoders. Bottom block: fine-tuned checkpoints. Sign of Δ is the central observation: zero-shot CLIP/SigLIP *gain* from blanking (negative confound removed); trained DINOv2 *loses* substantially (the model learned to depend on the title region).

Encoder	orig. R@10	b-15 R@10	Δ
CLIP zero-shot	0.779	0.852	+0.073
SigLIP zero-shot	0.624	0.731	+0.106
DINOv2 zero-shot	0.058	0.058	−0.000
I-JEPA zero-shot	0.015	0.012	−0.003
ImageNet zero-shot	0.031	0.032	+0.001
Plain-ViT zero-shot	0.029	0.030	+0.001
DINOv2-MLP-30k	0.787	0.494	−0.292
DINOv2-SALAD-30k	0.880	0.720	−0.161
DINOv2-SALAD-50k	0.915	0.673	−0.242
CLIP-SALAD-30k	0.724	0.687	−0.037

Table 13. Phase-2 cross-set R@1 with the top fraction of every anchor and pool image painted white. The same fine-tuned checkpoints that lose Protocol-A R@10 under blanking (Table 12) *gain* Phase-2 R@1 by the same information removal. CLIP+MLP-30k embeddings collapsed; all predictions are identical.

Run	orig. R@1	b-15 R@1	b-30 R@1
DINOv2-MLP-30k	0.169	0.712	0.763
DINOv2-SALAD-30k	0.212	0.686	0.737
DINOv2-SALAD-50k	0.110	0.661	0.737
CLIP-SALAD-30k	0.551	0.686	0.720
CLIP-MLP-30k†	0.932	0.932	0.932

† Collapsed model.